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Research Article

On the Heat and Wave Equations with the Sturm-Liouville Operator in Quantum Calculus

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In this paper, we explore a generalised solution of the Cauchy problems for the q-heat and q-wave equations which are generated by Jackson's and the q-Sturm-Liouville operators with respect to t and x, respectively. For this, we use a new method, where a crucial tool is used to represent functions in the Fourier series expansions in a Hilbert space on quantum calculus. We show that these solutions can be represented by explicit formulas generated by the q-Mittag-Leffler function. Moreover, we prove the unique existence and stability of the weak solutions.

1. Introduction

In the last decade, the theory of quantum groups and q-deformed algebras have been the subject of intense investigation. Many physical applications have been investigated on the basis of the q-deformation of the Heisenberg algebra (see [1, 2]). For instance, the q-deformed Schrödinger equations have been proposed in [3, 4], and applications to the study of q-deformed version of the hydrogen atom and of the quantum harmonic oscillator have been presented (see [5]). Fractional calculus and the q-deformed Lie algebras are closely related. A new class of fractional q-deformed Lie algebras is proposed, which for the first time allows a smooth transition between the different Lie algebras (see [6]).

The origin of the q-difference calculus can be traced back to the works by Jackson (see [7, 8]) and Carmichael (see [9]) from the beginning of the twentieth century, while basic definitions and properties can be found, e.g., in the monographs [10, 11] and the PhD thesis [12]. Recently, the fractional q-difference calculus has been proposed by Al-salam (see [13]) and Agarwal (see [14]). We can also mention papers [15, 16], where the authors investigated the explicit solutions to linear fractional q-differential equations with the q-fractional derivative, and in [17], the q-analogue nonhomogeneous wave equations were studied.

A motivation behind this work is to state some new results about the q-heat and q-wave equations associated to the q-Sturm-Liouville operator (see (10)). We attempt to extend the heat representation theory studied in some cases (see [18–20], etc.). We define a generalised solution of the Cauchy problem for these equations generated by the q-Mittag-Leffler function and the q-associated functions of a biorthogonal system (see (13)). We investigate the well-posedness of the Cauchy problem for the q-heat and q-wave equations for

operators with a discrete nonnegative spectrum acting on $L_q^2[0,1]$. In particular, we prove both unique existence and stability of the corresponding the generalised solution.

The paper is organized as follows: the main results are presented and proved in Section 3 and Section 4. In order to not disturb these presentations, we include in Section 2 some necessary Preliminaries.

2. Preliminaries

In this section, we recall some notations and basic facts in q-calculus. We will always assume that $0 < q < 1$. The q-real number $[\alpha]_q$ is defined by

$$[\alpha]_q = \frac{1-q^\alpha}{1-q}.$$

The q-shifted factorial is defined by

$$(a;q)_n = \begin{cases} 1, & n = 0, \\ (1-a)(1-aq)\cdots\left(1-aq^{n-1}\right), & n \in \mathbb{N}. \end{cases}$$

Moreover, their natural expansions to the reals are

$$(a-b)_q^\alpha = a^\alpha \frac{(b/a;q)_\infty}{(q^\alpha b/a;q)_\infty}, (a;q)_\alpha = \frac{(a;q)_\infty}{(aq^\alpha;q)_\infty}, (a;q)_\infty = \prod_{i=0}^{\infty}(1-aq^i). \tag{1}$$

The Jackson's q-difference operator $D_q f(x)$ is (see, [8, 12] Section 2.1])

$$D_q f(x) = \frac{f(x)-f(qx)}{x(1-q)}. \tag{2}$$

The q-derivative D_q of a product of the functions f and g as defined by

$$D_q(fg)(x) = f(qx)D_q(g)(x) + D_q(f)(x)g(x). \tag{3}$$

As given in [10], two q-analogues of the exponential functions are defined by

$$e_q^x = \frac{1}{((1-q)x;q)_\infty}, E_q^x = (-(1-q)x;q)_\infty. \tag{4}$$

Moreover, we have that

$$D_q e_q^x = e_q^x, D_q E_q^{-x} = E_q^{-qx}, e_q^x E_q^{-x} = 1. \tag{5}$$

Due to the various types of q-differences introduced in quantum calculus, trigonometric functions have various q-analogues (see, [21] Section 2 [10], Section 10 and [12], Section 2.12). The following definition of cosine and sine will be useful in this investigation (see [20]):

$$\cos\left(z;q^2\right) = \sum_{k=0}^{\infty}\frac{(-1)^k q^{k^2}z^{2k}}{[2k]_q!}, \sin\left(z;q^2\right) = \sum_{k=0}^{\infty}\frac{(-1)^k q^{k(k+1)}z^{2k+1}}{[2k+1]_q!}, \tag{6}$$

where the q-analogue of the binomial coefficients $[n]_q!$ is defined by

$$[n]_q! = \begin{cases} 1, & \text{if } n = 0, \\ [1]_q \times [2]_q \times \cdots \times [n]_q, & \text{if } n \in N. \end{cases}$$

The q-integral (or Jackson's integral) is defined by (see [8])

$$\int_0^x f(t)d_q t = (1-q)x\sum_{m=0}^{\infty} q^m f(xq^m), \tag{7}$$

and a more general form is given by

$$\int_a^b f(x)d_q x = \int_0^b f(x)d_q x - \int_0^a f(x)d_q x,$$

for $0 < a < b$.

The q-version of integration by parts reads

$$\int_a^b f(x)D_q g(x)d_q x = [fg]_a^b - \int_a^b g(qx)D_q f(x)d_q x, \tag{8}$$

and if $f \equiv 1$, then we get that

$$\int_a^b D_q g(x)d_q x = g(b) - g(a). \tag{9}$$

The q -Sturm-Liouville Problem. Let $L_q^2[0,1]$ be the space of all real-valued functions defined on $[0,1]$ such that

$$\|f\|_{L_q^2[0,1]} := \left(\int_0^1 |f(x)|^2 d_q x\right)^{1/2} < \infty.$$

The space $L_q^2[0,1]$ is a separable Hilbert space with the inner product:

$$\langle f, g\rangle := \int_0^1 f(x)g(x)d_q x, f, g \in L_q^2[0,1].$$

Now, we shortly describe the study introduced by Annaby and Mansour in of a basic q-Sturm-Liouville eigenvalue problem in a Hilbert space (see [21], Chapter 3). In particular, they investigated the basic q-Sturm-Liouville equation:

$$-\frac{1}{q}D_{q^{-1}}D_q y(x) + v(x)y(x) = \lambda y(x), (0 \le x \le 1; \lambda \in \mathbb{C}),$$

where $v(\cdot)$ is defined on $[0,1]$ and continuous at zero. Let $C_{q,0}^2[0,1]$ denotes the space of all functions $y(\cdot)$ such that y

and $D_q y$ are continuous at zero. If $\nu \equiv 0$, then we get the operator $\mathscr{L}$ in the following form:

$$\mathscr{L} := \begin{cases} -\frac{1}{q} D_{q^{-1}} D_q y(x) = \lambda y(x), \\ y(0) = y(1) = 0, \end{cases} \tag{10}$$

for $0 \le x \le 1$ and $\lambda \in \mathbb{R}$. The operator $\mathscr{L}$ is self adjoint on $C^2_{q,0}[0,1] \cap L^2_q[0,1]$ (see [21], Theorem 3.4.). A fundamental set of solutions of (10) are $\cos(\sqrt{\lambda}; q^2)$ and $\sin(\sqrt{\lambda}; q^2)/\sqrt{\lambda}$. Moreover, the eigenvalues $\{\lambda_k\}_{k=1}^{\infty}$ are the zeros of $\sin(\sqrt{\lambda_k}; q^2)$, where

$$\lambda_k = (1-q)^{-2} q^{-2k+2\mu_k^{-1/2}}, k = 0, 1, \cdots, \tag{11}$$

and $\sum_{k=1}^{\infty} \mu_k < \infty$, $0 \le \mu_k \le 1$, and

$$\lambda_0 := (1-q)^{-2} q \le \lambda_k, k = 1, 2, 3, \cdots. \tag{12}$$

Additionally, the corresponding set of eigenfunctions $\{\sin(\sqrt{\lambda_k}; q^2)/\sqrt{\lambda_k}\}_{k=1}^{\infty}$ is an orthogonal basis in $L^2_q(0,1)$. Thus, we can identify $f \in L^2_q[0,1]$ with its Fourier series:

$$f(x) := \sum_{k=1}^{\infty} \langle f, \phi_k \rangle \phi_k(x),$$

where

$$\phi_k(x) = \frac{\sin\left(\sqrt{\lambda_k} x; q^2\right)}{\sqrt{\lambda_k}}. \tag{13}$$

The Sobolev Space Associated with $\mathscr{L}$. The next step is to recall the essential elements of the Fourier analysis presented in [22–24], as well as its applications to the spectral properties of $\mathscr{L}$. The space $C^{\infty}_{\mathscr{L}}[0,1] := \bigcap_{m=1}^{\infty} \mathrm{Dom}(\mathscr{L}^m)$ is called the space of test functions for $\mathscr{L}$, where

$$\mathrm{Dom}(\mathscr{L}^m) := \left\{ f \in L^2_q[0,1] : \mathscr{L}^j f \in \mathrm{Dom}(\mathscr{L}), j = 0, 1, 2, \cdots, m-1 \right\}.$$

For $g \in C^{\infty}_{\mathscr{L}}[0,1]$, we introduce the Fréchet topology of $C^{\infty}_{\mathscr{L}}[0,1]$ by the family of norms:

$$\|g\|_{C^m_{\mathscr{L}}[0,1]} := \max_{i \le m} \left\| \mathscr{L}^i g \right\|_{L^2_q[0,1]}.$$

The space of $\mathscr{L}$-distributions $\mathscr{D}'_{\mathscr{L}}[0,1] := L(C^{\infty}_{\mathscr{L}}[0,1], \mathbb{R})$ is the space of all linear continuous functionals on $C^{\infty}_{\mathscr{L}}[0,1]$.

Thus, for $s \in \mathbb{R}$, we can also define the Sobolev spaces $W^s_{q,\mathscr{L}}$ associated to $\mathscr{L}$ in the following form:

$$W^s_{q,\mathscr{L}} := \left\{ f \in \mathscr{D}'_{\mathscr{L}}[0,1] : \mathscr{L}^{s/2} f \in L^2_q[0,1] \right\},$$

with the norm $\|f\|_{W^s_{q,\mathscr{L}}} := \|\mathscr{L}^{s/2} f\|_{L^2_q[0,1]}$.

For $m \in \mathbb{N}_0$, we introduce the space $C^m_q([0,1]; W^s_{q,\mathscr{L}}[0,1])$ defined by the norms

$$\|u\|_{C^m_q([0,T]; W^s_{q,\mathscr{L}}[0,1])} := \sum_{n=0}^{m} \max_{0 \le t \le T} \left\| D^n_{q,t} u(t,.) \right\|_{W^s_{q,\mathscr{L}}[0,1]}, 0 < T < \infty,$$

where the q-partial differential operator $D_{q,t} u(t,x)$ with respect to t has the following form:

$$D_{q,t} u(t,x) = \frac{u(t,x) - u(qt,x)}{(1-q)t}.$$

Notation: the symbol $M \lesssim K$ means that there exists $\gamma > 0$ such that $M \le \gamma K$, where γ is a constant.

3. The q-Heat Equation

We start with a study of the following Cauchy problem:

$$D_{q,t} u(t,x) + \mathscr{L} u(t,x) = f(t,x), x \in [0,1], t > 0, \tag{14}$$

with the initial condition

$$u(0,x) = \varphi(x), x \in [0,1]. \tag{15}$$

We say a generalised solution of the problem (14)-(15) is a function $u(t,x)$ such that they satisfy equation (14) and condition (15).

Theorem 1. *We assume that $0 < T < \infty$. Let $\varphi \in W^2_{q,\mathscr{L}}[0,1]$ and $f \in C([0,T]; W^2_{q,\mathscr{L}}[0,1])$. Then, there exists the generalised solution of u to problem (14)-(15), and*

$$u \in C^1_q\left([0,T]; L^2_q[0,1]\right) \cap C\left([0,T]; W^2_{q,\mathscr{L}}[0,1]\right). \tag{16}$$

Moreover, this solution can be written in the following explicit form

$$u(t,x) = \sum_{k \in \mathbb{N}} \left[e^{-\lambda_k t}_q \varphi_k + e^{-\lambda_k t}_q \int_0^t E^{\lambda_k qs}_q f_k(s,\cdot) d_q s \right] \phi_k(x). \tag{17}$$

Proof. Existence. Since the system of eigenfunctions $\{\phi_k\}_{k=1}^{\infty}$ is a basis in $L^2_q[0,1]$ (see (11)), we seek for a function $u(t,x)$ in the form

$$u(t,x) = \sum_{k \in \mathbb{N}} u_k(t) \phi_k(x), \tag{18}$$

for each fixed $0 < t < T < \infty$. The coefficients will then be given by the Fourier coefficients formula $u_k(t) = \langle u(t,\cdot)\phi_k\rangle$.

We can similarly expand the source function,

$$f(t,x) = \sum_{k\in\mathbb{N}} f_k(t)\phi_k(x), f_k(t) = \langle f(t,\cdot),\phi_k\rangle. \tag{19}$$

From (11) and (18), we have that

$$\mathscr{L}\phi_k(x) = \lambda_k\phi_k(x), k\in\mathbb{N}.$$

Hence,

$$\mathscr{L}u(t,x) = \sum_{k\in\mathbb{N}_0} u_k(t)\lambda_k\phi_k(x), \tag{20}$$

and

$$D_{q,t}u(t,x) = \sum_{k\in\mathbb{N}} D_q u_k(t)\phi_k(x). \tag{21}$$

Substituting (20) and (21) into the equation (14), we find that

$$\sum_{k\in\mathbb{N}}\left[D_q u_k(t) + \lambda_k u_k(t)\right]\phi_k(x) = \sum_{k\in\mathbb{N}} f_k(t)\phi_k(x). \tag{22}$$

But then, due to the completeness,

$$D_q u_k(t) + \lambda_k u_k(t) = f_k(t), k\in\mathbb{N}, \tag{23}$$

which are ODEs for the coefficients $u_k(t)$ of the series (18). Using the integrating factor $E_q^{\lambda_k qt}$ and (2) and (3), we can rewrite the ODE as

$$\begin{aligned} E_q^{\lambda_k qt} f_k(t) &= E_q^{\lambda_k qt} D_q u_k(t) + E_q^{\lambda_k qt}\lambda_k u_k(t) \\ &= E_q^{\lambda_k qt} D_q u_k(t) + D_q\left[E_q^{\lambda_k t}\right] u_k(t) \\ &= D_q\left[E_q^{\lambda_k t} u_k(t)\right]. \end{aligned} \tag{24}$$

Form (3), (5), and (24), we get that

$$\int_0^t D_q\left[E_q^{\lambda_k t} u_k(t)\right] d_q s = \int_0^t E_q^{\lambda_k qs} f_k(t) f_k(s) d_q s,$$

so that

$$E_q^{\lambda_k t} u_k(t) = u_k(0) + \int_0^t E_q^{\lambda_k qt} f_k(s) d_q s,$$

which, in its turn, implies that

$$u_k(t) = \frac{u_k(0)}{E_q^{\lambda_k t}} + \frac{1}{E_q^{\lambda_k t}}\int_0^t E_q^{\lambda_k qs} f_k(s) d_q s,$$

and we conclude that

$$u_k(t) = e_q^{-\lambda_k t} u_k(0) + e_q^{-\lambda_k t}\int_0^t E_q^{\lambda_k qs} f_k(s) d_q s.$$

But the initial conditions (16) and (22) imply that $u_k(0) = \varphi_k$. Thus,

$$u_k(t) = e_q^{-\lambda_k t}\varphi_k + e_q^{-\lambda_k t}\int_0^t E_q^{\lambda_k qs} f_k(s,\cdot) d_q s. \tag{25}$$

Therefore, the solution $u(t,x)$ can be written in the series form as

$$u(t,x) = \sum_{k\in\mathbb{N}}\left[e_q^{-\lambda_k t}\varphi_k + e_q^{-\lambda_k t}\int_0^t E_q^{\lambda_k qs} f_k(s,\cdot) d_q s\right]\phi_k(x),$$

so, also (17) is proved.

Convergence. From (1), (4), and (5), we have that

$$e_q^{-x} = \frac{1}{(-(1-q)x;q)_\infty} \le \frac{1}{1+(1-q)x} \le 1, E_q^{qx} \le E_q^x,$$

for $x\in[0,1]$. Hence, using for $0 < t < T < \infty$, (5), (23), and (25), we get that

$$\begin{aligned} |u_k(t)| &\overset{(25)}{\le} e_q^{-\lambda_k t}|\varphi_k| + \int_0^t \frac{E_q^{\lambda_k qs}}{e_q^{\lambda_k t}}|f_k(s)| d_q s \le |\langle\varphi,\phi_k\rangle| \\ &+ \int_0^t |\langle f(s,\cdot),\phi_k\rangle| d_q s \le |\langle\varphi,\phi_k\rangle| \\ &+ T\max_{0\le s\le T}|\langle f(s,\cdot),\phi_k\rangle| \le \max\{1,T\} \\ &\cdot\left[|\langle\varphi,\phi_k\rangle| + \max_{0\le s\le T}|\langle f(s,\cdot),\phi_k\rangle|\right] \precsim |\langle\varphi,\phi_k\rangle| \\ &+ \max_{0\le s\le T}|\langle f(s,\cdot),\phi_k\rangle|, \end{aligned} \tag{26}$$

and

$$\begin{aligned} \left|D_q u_k(t)\right| &\overset{(23)}{\le} \lambda_k|u_k(t)| + |f_k(t)| \overset{(26)}{\precsim} |\langle\lambda_k\varphi,\phi_k\rangle| \\ &+ |\langle\lambda_k f_k(t,\cdot),\phi_k\rangle| + \lambda_k^{-1}|\langle\lambda_k f_k(t,\cdot),\phi_k\rangle| \le |\langle\lambda_k\varphi,\phi_k\rangle| \\ &+ (1+\lambda_0)|\langle\lambda_k f_k(t,\cdot),\phi_k\rangle| \precsim |\langle\mathscr{L}\varphi,\phi_k\rangle| \\ &+ \max_{0\le t\le T}|\langle\mathscr{L} f_k(t,\cdot),\phi_k\rangle|. \end{aligned} \tag{27}$$

Hence,

$$\begin{aligned} |\mathscr{L}u(t,\cdot)| &= |\langle\lambda_k u_k(t),\phi_k\rangle| \overset{(26)}{\precsim} |\langle\lambda_k\varphi,\phi_k\rangle| \\ &+ \max_{0\le s\le T}|\langle\lambda_k f(s,\cdot),\phi_k\rangle| = |\langle\mathscr{L}\varphi,\phi_k\rangle| \\ &+ \max_{0\le s\le T}|\langle\mathscr{L} f(s,\cdot),\phi_k\rangle|. \end{aligned} \tag{28}$$

Since $\varphi\in W^2_{q,\mathscr{L}}$, $f\in C([0,1]; W^2_{q,\mathscr{L}})$, and, hence, by using the Plancherel identity and (27) and (28), we can conclude

that

$$\|u(t,\cdot)\|^2_{L^2_q[0,1]} = \sum_{k\in\mathbb{N}} |u_k(t)|^2 \overset{(3.13)}{\lesssim} \sum_{k\in\mathbb{N}} |\langle\varphi,\phi_k\rangle|^2 + \max_{0\le s\le T}\sum_{k\in\mathbb{N}} |\langle f(s,\cdot),\phi_k\rangle|^2 = \|\varphi\|^2_{W^2_{q,\mathscr{L}}[0,1]} + \|f\|^2_{C\left([0,T];W^2_{q,\mathscr{L}}[0,1]\right)} < \infty,$$

and

$$\left\|D_q u(t,\cdot)\right\|^2_{L^2_q[0,1]} = \sum_{k\in\mathbb{N}} \left|D_q u_k(t,\cdot)\right|^2 \overset{(27)}{\lesssim} \sum_{k\in\mathbb{N}} |\langle\mathscr{L}\varphi,\phi_k\rangle|^2 + \max_{0\le s\le T}\sum_{k\in\mathbb{N}} |\langle\mathscr{L}f(s,\cdot),\phi_k\rangle|^2 = \|\varphi\|^2_{W^2_{q,\mathscr{L}}[0,1]} + \|f\|^2_{C\left([0,T];W^2_{q,\mathscr{L}}[0,1]\right)} < \infty,$$

and

$$\|\mathscr{L}u(t,\cdot)\|^2_{L^2_q[0,1]} \overset{(28)}{\lesssim} \|\varphi\|^2_{W^2_{q,\mathscr{L}}[0,1]} + \|f\|^2_{C\left([0,T];W^2_{q,\mathscr{L}}[0,1]\right)} < \infty,$$

which mean that $u \in C^1_q([0,T]; L^2_q[0,1]) \cap C([0,T]; W^2_{q,\mathscr{L}}[0,1])$.

Uniqueness. It only remains to prove the uniqueness of the solution. We assume the opposite; namely, that there exist functions $u(t,x)$ *and* $v(t,x)$*, which are two different solutions of problem (14)-(15). Let* $0 < t < T < \infty$*. Then, we have that*

$$\begin{cases} D_{q,t}u(t,x) + \mathscr{L}u(t,x) = f(t,x), & 0 < x < 1,\\ u(0,x) = \varphi(x), & 0 \le x \le 1,\end{cases}$$
$$\begin{cases} D_{q,t}v(t,x) + \mathscr{L}v(t,x) = f(t,x), & 0 < x < 1,\\ v(0,x) = \varphi(x), & 0 \le x \le 1.\end{cases}$$

We define $W(t,x) = u(t,x) - v(t,x)$*. Then, the function* $W(t,x)$ *is a solution of the following problem*

$$\begin{cases} D_{q,t}w(t,x) + \mathscr{L}w(t,x) = 0, & 0 < x < 1,\\ w(0,x) = 0, & 0 \le x \le 1.\end{cases}$$

From (18), it follows that $W(t,x) \equiv 0$*, that is,* $u(x,t) \equiv v(x,t)$*, and this contradiction to our assumption proves the uniqueness of the solution. The proof is complete.*

4. The q-Wave Equation

In this section, we will seek for a generalised function $u(t,x)$, which satisfies the following q-wave equation

$$D^2_{q,t}u(t,x) + \mathscr{L}u(t,x) = f(t,x), 0 < x < 1, \tag{29}$$

for $0 < t < T < \infty$ with the initial conditions

$$u(0,x) = \psi(x), D_{q,t}u(0,x) = \eta(x), 0 < x \le 1. \tag{30}$$

Theorem 2. *We assume that* $0 < T < \infty$*. Let* $\psi, \eta \in W^2_{q,\mathscr{L}}[0,1]$ *and* $f \in C^1_q([0,T]; W^2_{q,\mathscr{L}}[0,1])$*. Then, there exists the generalised solution of problem (29)-(30):*

$$u \in C^2_q\left([0,1]; L^2_q[0,T]\right) \cap C\left([0,T]; W^2_{q,\mathscr{L}}[0,1]\right).$$

Moreover, this solution can be written in the following explicit form:

$$u(t,x) = \sum_{k\in\mathbb{N}_0}\Bigg(\psi_k e_{2,1}\left(-\lambda_k t^2; q\right) + t\eta_k e_{2,2}\left(\lambda_k t^2; q\right) - \frac{1}{\lambda_k}f_k(0)e_{2,1}\left(-\lambda_k t^2; q\right) - \frac{1}{\lambda_k}\int_0^t e_{2,1} \cdot \left(\lambda_k\left(t - q^3 s\right)^2_q; q\right) D_{q,s}f_k(s)d_q s\Bigg)\phi_k, \tag{31}$$

where the q-Mittag-Leffler function $e_{\alpha,\beta}(\lambda_k(t-q^2s)^\alpha_q; q)$ *is given by (see [25] and [26], Section 7):*

$$e_{\alpha,\beta}\left(\lambda_k(t-qs)^\alpha_q; q\right) = \sum_{m=0}^{\infty}\frac{\lambda^m_k(t-qs)^{m\alpha}_q}{\Gamma_q(m\alpha+\beta)}, \tag{32}$$

for $\alpha, \beta \in \mathbb{R}$ *and* $0 < s \le t < \infty$*, where the gamma function* $\Gamma_q(x)$ *is defined by*

$$\Gamma_q(x) = \frac{(q,q)^\infty_q}{(q^x,q)^\infty_q}(1-q)^{1-x}, \Gamma_q(n+1) = [n]_q!, n \in \mathbb{N}. \tag{33}$$

Proof. Existence. By repeating the arguments in the proof of Theorem 1., we have the Cauchy type problem:

$$D^2_q u_k(t) + \lambda_k u_k(t) = f_k(t), k \in \mathbb{N}_0, \tag{34}$$

with the initial conditions

$$u_k(0) = \psi_k, D_q u_k(0) = \eta_k, k \in \mathbb{N}_0, \tag{35}$$

where $f_k(t) = \langle f(t,\cdot)\phi_k\rangle$*,* $\psi_k = \langle\psi(\cdot)\phi_k\rangle$ *and* $\eta_k = \langle\eta(\cdot)\phi_k\rangle$*.*

Then, the solution to this Cauchy type in problem (29)-(30) is given (see [25], Example 6)

$$u_k(t) = \psi_k e_{2,1}\left(-\lambda_k t^2; q\right) + t\eta_k e_{2,2}\left(-\lambda_k t^2; q\right) + \int_0^t (t-qs)e_{2,2}\left(-\lambda_k\left(t-q^2s\right)^2_q; q\right)f_k(s)d_q s. \tag{36}$$

By using (2) and we find that

$$\begin{aligned} D_{q,s}\left[e_{2,1}\left(-\lambda_k\left(t-q^2s\right)_q^2;q\right)\right] &= -\sum_{k\in\mathbb{N}}\frac{(-\lambda_k)^m}{\Gamma_q(2m+1)}[2m]_q\left(t-q^3s\right)_q^{2m-1} \\ &= \lambda_k(t-qs)\sum_{k\in\mathbb{N}}\frac{(-\lambda_k)^{m-1}}{\Gamma_q(2m)}\left(t-q^2s\right)_q^{2m-2} \\ &= \lambda_k(t-qs)e_{2,2}\left(\lambda_k\left(t-q^2s\right)_q^2;q\right). \end{aligned} \tag{37}$$

By applying (8) and using (36) and (37), we get that

$$\begin{aligned} u_k(t) &= \psi_k e_{2,1}\left(-\lambda_k t^2;q\right)+t\eta_k e_{2,2}\left(-\lambda_k t^2;q\right) \\ &\quad+\frac{1}{\lambda_k}\int_0^t D_{q,s}\left[e_{2,1}\left(-\lambda_k\left(t-q^2s\right)_q^2;q\right)\right]f_k(s)d_qs \\ &= \psi_k e_{2,1}\left(-\lambda_k t^2;q\right)+t\eta_k e_{2,2}\left(-\lambda_k t^2;q\right) \\ &\quad+\frac{1}{\lambda_k}f_k(t)e_{2,1}\left(-\lambda_k\left(t-q^2\right)_q^2;q\right) \\ &\quad-\frac{1}{\lambda_k}f_k(0)e_{2,1}\left(-\lambda_k t^2;q\right) \\ &\quad-\frac{1}{\lambda_k}\int_0^t e_{2,1}\left(-\lambda_k\left(t-q^3s\right)_q^2;q\right)D_{q,s}f_k(s)d_qs. \end{aligned} \tag{38}$$

Since $e_{2,1}(-\lambda_k t^2(q^2;q)_2)\equiv 0$ (see [21], Theorem 7.12]), by using (18) and (38), it follows that solution exists and can be written as

$$\begin{aligned} u(t,x) = \sum_{k\in\mathbb{N}}\Bigg(&\psi_k e_{2,1}\left(-\lambda_k t^2;q\right)+t\eta_k e_{2,2}\left(\lambda_k t^2;q\right) \\ &-\frac{1}{\lambda_k}f_k(0)e_{2,1}\left(-\lambda_k t^2;q\right)-\frac{1}{\lambda_k}\int_0^t e_{2,1} \\ &\cdot\left(-\lambda_k\left(t-q^3s\right)_q^2;q\right)D_{q,s}f_k(s)d_qs\Bigg)\phi_k, \end{aligned}$$

i.e., on the explicit form (34).

Convergence. Firstly, using the results in [27], Lemma 6 and in [17], Lemma 1 for the q-trigonometric functions in (6), we see that $e_{2,2}(-\lambda_k t^2;q)$ and $e_{2,1}(-\lambda_k t^2;q)$ are also bounded with $t>0$. Then, forms (4), (12), and (32) follow that

$$\begin{aligned} \left|e_{2,2}\left(-\lambda_k t^2;q\right)\right| &= \left|\sum_{m=0}^{\infty}\frac{(-\lambda_k)^m t^{2m}}{[2m+1]_q!}\right| \\ &\le \sum_{m=0}^{\infty}\frac{\lambda_k^m T^{2m}}{[2m+1]_q!} \\ &= \frac{\sin\left(\sqrt{\lambda_k}T;q^2\right)}{2T\sqrt{\lambda_k}} \\ &\le \frac{C_{1,q}}{2T\sqrt{\lambda_k}}, \end{aligned} \tag{39}$$

and

$$\begin{aligned} \left|e_{2,1}\left(-\lambda_k t^2;q\right)\right| &= \left|\sum_{m=0}^{\infty}\frac{(-1)^m\left(T\sqrt{\lambda_k}\right)^{2m}}{[2m]_q!}\right| \\ &\le \cos\left(\sqrt{\lambda_k}T;q^2\right)\le C_{2,q}, \end{aligned} \tag{40}$$

where $C_{1,q}, C_{2,q}$ are any constant which only depends on q.

Next, by using (38), (39), and (40), we obtain that

$$\begin{aligned} |u_k(t)| &\overset{(40)}{\lesssim} |\langle\psi,\phi_k\rangle|+|\langle\eta,\phi_k\rangle|+\frac{|\langle f(0,\cdot),\phi_k\rangle|}{\lambda_k} \\ &\quad+\frac{1}{\lambda_k}\int_0^t\left|\left\langle D_q f(s,\cdot),\phi_k\right\rangle\right|d_qs\le|\langle\psi,\phi_k\rangle|+|\langle\eta,\phi_k\rangle| \\ &\quad+\frac{|\langle f(0,\cdot),\phi_k\rangle|}{\lambda_k}+\frac{T}{\lambda_k}\max_{0\le s\le T}\left|\left\langle D_q f(s,\cdot),\phi_k\right\rangle\right|\le C_{\lambda_0} \\ &\quad\cdot\left[|\langle\psi,\phi_k\rangle|+|\langle\eta,\phi_k\rangle|+\sum_{m=0}^{1}\max_{0\le t\le T}\left|\left\langle D_q^m f(t,\cdot),\phi_k\right\rangle\right|\right], \end{aligned} \tag{41}$$

where $C_{\lambda_0}:=\max\{1,(1/\lambda_0),(T/\lambda_0)\}$.

Therefore, by using (7), (8), (30), (34), (33), and (41), we have that

$$\begin{aligned} \left|D_q u_k(t)\right| &= \left|-D_q u_k(0)+\int_0^t D_q^2 u_k(s)d_qs\right| \overset{(30)(34)}{\le} |\langle\eta,\phi_k\rangle| \\ &\quad+\int_0^t|\langle f(s,\cdot),\phi_k\rangle|d_qs \\ &\quad+\lambda_k\int_0^t|u_k(s)|d_qs\overset{(41)}{\lesssim}\frac{1}{\lambda_k}|\langle\lambda_k\eta,\phi_k\rangle| \\ &\quad+\frac{T}{\lambda_k}\max_{0\le s\le T}|\langle\lambda_k f(s,\cdot),\phi_k\rangle|+TC_{\lambda_0} \\ &\quad\cdot\left[|\langle\lambda_k\psi,\phi_k\rangle|+|\langle\lambda_k\eta,\phi_k\rangle|+\sum_{m=0}^{1}\max_{0\le t\le T}\left|\left\langle\lambda_k D_q^m f(t,\cdot),\phi_k\right\rangle\right|\right] \\ &\lesssim|\langle\mathscr{L}\psi,\phi_k\rangle|+|\langle\mathscr{L}\eta,\phi_k\rangle|+\sum_{m=0}^{1}\max_{0\le t\le T}\left|\left\langle\mathscr{L}D_q^m f(t,\cdot),\phi_k\right\rangle\right|, \end{aligned} \tag{42}$$

and

$$\begin{aligned} \left|D_a^2 u_k(t)\right| &\overset{(34)}{\lesssim}\lambda_k|u_k(t)|+|f_k(t)|\overset{(41)}{\lesssim}|\langle\lambda_k\psi,\phi_k\rangle|+|\langle\lambda_k\eta,\phi_k\rangle| \\ &\quad+\sum_{m=0}^{1}\max_{0\le t\le T}\left|\left\langle\lambda_k D_q^m f(t,\cdot),\phi_k\right\rangle\right|=|\langle\mathscr{L}\psi,\phi_k\rangle|+|\langle\mathscr{L}\eta,\phi_k\rangle| \\ &\quad+\sum_{m=0}^{1}\max_{0\le t\le T}\left|\left\langle\mathscr{L}D_q^m f(t,\cdot),\phi_k\right\rangle\right|. \end{aligned} \tag{43}$$

Thus,

$$\begin{aligned}\|u(t)\|^2_{L^2_q[0,1]} &= \sum_{k\in\mathbb{N}} |u_k(t)|^2 \overset{(41)}{\lesssim} \sum_{k\in\mathbb{N}} |\langle \psi, \phi_k\rangle|^2 + \sum_{k\in\mathbb{N}} |\langle \eta, \phi_k\rangle|^2 \\ &\quad + \sum_{m=0}^{1} \max_{0\le t\le T} \sum_{k\in\mathbb{N}} \left| \left\langle D_q^m f(t,\cdot), \phi_k \right\rangle \right|^2 \\ &= \|\eta\|^2_{W^2_{q,\mathscr{L}}[0,1]} + \|\psi\|^2_{W^2_{q,\mathscr{L}}[0,1]} \\ &\quad + \|f\|^2_{C^1_q\left([0,T];W^2_{q,\mathscr{L}}[0,1]\right)} < \infty,\end{aligned}$$

$$\begin{aligned}\left\|D_q u(t)\right\|^2_{L^2_q[0,1]} &= \sum_{k\in\mathbb{N}} \left|D_q u_k(t)\right|^2 \lesssim \sum_{k\in\mathbb{N}} |\langle \mathscr{L}\psi, \phi_k\rangle|^2 \\ &\quad + \sum_{k\in\mathbb{N}} |\langle \mathscr{L}\eta, \phi_k\rangle|^2 \\ &\quad + \sum_{m=0}^{1} \max_{0\le t\le T} \left| \sum_{k\in\mathbb{N}} \left\langle \mathscr{L} D_q^m f(t,\cdot), \phi_k \right\rangle \right|^2 \\ &\le \|\eta\|^2_{W^2_{q,\mathscr{L}}[0,1]} + \|\psi\|^2_{W^2_{q,\mathscr{L}}[0,1]} \\ &\quad + \|f\|^2_{C^1_q\left([0,T];W^2_{q,\mathscr{L}}[0,1]\right)} < \infty,\end{aligned}$$

and

$$\begin{aligned}\left\|D_q^2 u(t)\right\|^2_{L^2_q[0,1]} &= \sum_{k\in\mathbb{N}} \left|D_q^2 u_k(t)\right|^2 \lesssim \|\eta\|^2_{W^2_{q,\mathscr{L}}[0,1]} \\ &\quad + \|\psi\|^2_{W^2_{q,\mathscr{L}}[0,1]} + \|f\|^2_{C^1_q\left([0,T];W^2_{q,\mathscr{L}}[0,1]\right)} < \infty,\end{aligned}$$

and

$$\begin{aligned}\|\mathscr{L}u(t)\|^2_H &= \sum_{k\in I} \left|\langle \mathscr{L}u(t), \phi_k\rangle_H\right|^2 \\ &= \sum_{k\in I} [\lambda_k |u_k(t)|]^2 \lesssim \|\eta\|^2_{W^2_{q,\mathscr{L}}[0,1]} \\ &\quad + \|\psi\|^2_{W^2_{q,\mathscr{L}}[0,1]} \\ &\quad + \|f\|^2_{C^1_q\left([0,T];W^2_{q,\mathscr{L}}[0,1]\right)},\end{aligned}$$

which is means that $u \in C^2_q([0,1]; L^2_q[0,T]) \cap C_q([0,T]; W^2_{q,\mathscr{L}}[0,1])$.

Uniqueness. This part can be proved completely similar as the proof of Theorem 1.. So we omit the details.

Data Availability

Data supporting this manuscript are available from Scopus, Web of Science, and Google Scholar.

Conflicts of Interest

The authors declare that they have no conflicts of interest.

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